

Chapter 11 Part 3: pH Calculations and Buffers

The ionization of cyanic acid, HOCN , is represented above. A certain solution of HOCN has a pH of 2.0. What is the approximate concentration of unionized $\text{HOCN}(aq)$ in the solution?

- (A) $3 \times 10^{-4} M$
- (B) $1 \times 10^{-2} M$
- (C) $3 \times 10^{-1} M$
- (D) $2 \times 10^0 M$



The dissociation of the weak acid HF in water is represented by the equation above. Adding a 1.0 mL sample of which of the following would increase the percent ionization of $\text{HF}(aq)$ in 10 mL of a solution of 1.0 M HF?

- (A) 1.0 M KF
- (B) 1.0 M H_2SO_4
- (C) 10.0 M HF
- (D) Distilled water

3. Caffeine ($\text{C}_8\text{H}_{10}\text{N}_4\text{O}_2$) is a weak base with a K_b value of 4×10^{-4} . The pH of a 0.01 M solution of caffeine is in the range of

- (A) 2–3
- (B) 5–6
- (C) 7–8
- (D) 11–12

4. The pH of 0.1-molar ammonia is approximately

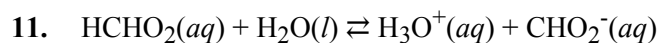
- (A) 1
- (B) 4
- (C) 7
- (D) 11
- (E) 14

5. The pH of a 0.01 M $\text{HNO}_2(aq)$ solution is in which of the following ranges? (For $\text{HNO}_2(aq)$, $K_a = 4.0 \times 10^{-4}$)

- (A) Between 1 and 2
- (B) Between 2 and 3
- (C) Between 4 and 5
- (D) Between 6 and 7

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6. What is the pH of a 1.0×10^{-2} -molar solution of HCN? (For HCN, $K_a = 4.0 \times 10^{-10}$.)
- (A) 10
(B) Between 7 and 10
(C) 7
(D) Between 4 and 7
(E) 4
7. What is the $\text{H}^+(aq)$ concentration in 0.05 M HCN(aq)? (The K_a for HCN is 5.0×10^{-10} .)
- (A) $2.5 \times 10^{-11} M$
(B) $2.5 \times 10^{-10} M$
(C) $5.0 \times 10^{-10} M$
(D) $5.0 \times 10^{-6} M$
(E) $5.0 \times 10^{-4} M$
8. A 0.10 M solution of a weak monoprotic acid has a pH equal to 4.0. The ionization constant, K_a , of the acid is
- (A) 1×10^{-3}
(B) 1×10^{-4}
(C) 1×10^{-7}
(D) 1×10^{-8}
(E) 1×10^{-9}
9. $\text{HNO}_2(aq) \rightleftharpoons \text{H}^+(aq) + \text{NO}_2^-(aq) K_a = 4.0 \times 10^{-4}$
- On the basis of the information above, what is the approximate percent ionization of HNO_2 in a 1.0 M $\text{HNO}_2(aq)$ solution?
- (A) 0.00040%
(B) 0.020%
(C) 0.040%
(D) 2.0%
10. $\text{HNO}_2(aq) \rightleftharpoons \text{H}^+(aq) + \text{NO}_2^-(aq) K_a = 4.5 \times 10^{-4}$
- On the basis of the information above, what is the approximate percent ionization of HNO_2 in a 1.0 M $\text{HNO}_2(aq)$ solution?
- (A) 0.00045%
(B) 0.021%
(C) 0.045%
(D) 2.1%

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$\text{HCHO}_2(aq)$, a weak acid ($K_a = 2 \times 10^{-4}$), dissociates in water according to the equation above. Which of the following provides the best estimate of the pH of 0.5 M $\text{HCHO}_2(aq)$ and identifies the species at the highest concentration (excluding H_2O) in the solution?

pH:

Species at Highest Concentration:

- (A) $\overset{1}{\text{H}_3\text{O}^+(aq)}$
- (B) $\overset{1}{\text{HCHO}_2(aq)}$
- (C) $\overset{2}{\text{H}_3\text{O}^+(aq)}$
- (D) $\overset{2}{\text{HCHO}_2(aq)}$

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12. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to CO_2 .

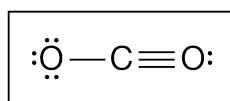


Diagram X

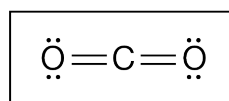
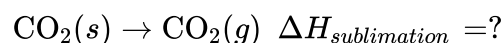
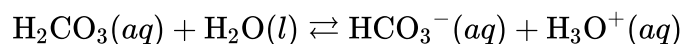


Diagram Z

- (a) Two possible Lewis electron-dot diagrams for CO_2 are shown above. Explain in terms of formal charges why diagram Z is the better diagram.
- (b) Identify the hybridization of the valence orbitals of the C atom in the CO_2 molecule represented in diagram Z.
- (c) A 0.1931 mol sample of dry ice, $\text{CO}_2(s)$, is added to an empty balloon. After the balloon is sealed, the $\text{CO}_2(s)$ sublimates and the $\text{CO}_2(g)$ in the balloon eventually reaches a temperature of 21.0°C and pressure of 0.998 atm. The physical change is represented by the following equation.

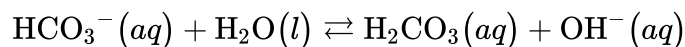


- (i) What is the sign (positive or negative) of the enthalpy change for the process of sublimation? Justify your answer.
- (ii) List all the numerical values of the quantities, with appropriate units, that are needed to calculate the volume of the balloon.
- (iii) Calculate the final volume, in liters, of the balloon.
- (d) $\text{CO}_2(g)$ reacts with water to form $\text{H}_2\text{CO}_3(aq)$, which quickly ionizes in water to form $\text{HCO}_3^-(aq)$ and $\text{H}_3\text{O}^+(aq)$, as shown in the following equation.



- (i) Identify one of the conjugate acid-base pairs in the equation above.
- (ii) When the concentration of CO_2 in water increases, more H_2CO_3 forms. Does the increase in the formation of H_2CO_3 cause the value of the reaction quotient, Q , for the system to increase, decrease, or remain the same? Justify your answer.
- (e) A student investigates the acid-base properties of $\text{HCO}_3^-(aq)$. The reaction of $\text{HCO}_3^-(aq)$ with $\text{H}_2\text{O}(l)$ is represented below.

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(i) The student measured the pH of a 0.100 *M* solution of $\text{HCO}_3^-(aq)$ as 9.68. Calculate the concentration of $\text{OH}^-(aq)$ in the solution.

$$K_b = \frac{[\text{H}_2\text{CO}_3][\text{OH}^-]}{[\text{HCO}_3^-]}$$

(ii) The K_b expression for the reaction of $\text{HCO}_3^-(aq)$ with $\text{H}_2\text{O}(l)$ is given above. Calculate the value of K_b for the reaction.

(iii) The student then carried out a second experiment at the same temperature to determine K_b . In the experiment, the student measured the pH of 0.300 *M* $\text{HCO}_3^-(aq)$. Would the value of K_b calculated in the second experiment be greater than, less than, or equal to the value of K_b calculated in part (e)(ii)? Justify your answer.

Refer to the following.

Concentration (<i>M</i>)	pH of Acid 1	pH of Acid 2	pH of Acid 3	pH of Acid 4
0.010	3.44	2.00	2.92	2.20
0.050	3.09	1.30	2.58	1.73
0.10	2.94	1.00	2.42	1.55
0.50	2.69	0.30	2.08	1.16
1.00	2.44	0.00	1.92	0.98

The pH of solutions of four acids prepared at various concentrations were measured and recorded in the table above.

The four acids are, in no particular order, chlorous, hydrochloric, lactic, and propanoic.

13. A 25 mL sample of a 1.0 *M* solution of acid 1 is mixed with 25 mL of 0.50 *M* NaOH. Which of the following best explains what happens to the pH of the mixture when a few drops of 1.0 *M* HNO_3 are added?
- (A) The pH of the mixture increases sharply, because HNO_3 is a strong acid.
- (B) The pH of the mixture decreases sharply, because H_3O^+ ions were added.
- (C) The pH of the mixture stays about the same, because the conjugate base of acid 1 reacts with the added H_3O^+ ions.
- (D) The pH of the mixture stays about the same, because the OH^- ions in the solution react with the added H_3O^+ ions.
14. A solution containing HCl and the weak acid HClO_2 has a pH of 2.4. Enough $\text{KOH}(aq)$ is added to the solution to increase the pH to 10.5. The amount of which of the following species increases as the $\text{KOH}(aq)$ is added?

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- (A) $\text{Cl}^-(aq)$
(B) $\text{H}^+(aq)$
(C) $\text{ClO}_2^-(aq)$
(D) $\text{HClO}_2(aq)$
15. A solution is prepared by adding 100 mL of 1.0 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ to 100 mL of 1.0 M $\text{NaC}_2\text{H}_3\text{O}_2(aq)$. The solution is stirred and its pH is measured to be 4.73. After 3 drops of 1.0 M HCl are added to the solution, the pH of the solution is measured and is still 4.73. Which of the following equations represents the chemical reaction that accounts for the fact that acid was added but there was no detectable change in pH?
- (A) $\text{H}_3\text{O}^+(aq) + \text{OH}^-(aq) \rightarrow 2 \text{H}_2\text{O}(l)$
(B) $\text{H}_3\text{O}^+(aq) + \text{Cl}^-(aq) \rightarrow \text{HCl}(g) + \text{H}_2\text{O}(l)$
(C) $\text{H}_3\text{O}^+(aq) + \text{C}_2\text{H}_3\text{O}_2^-(aq) \rightarrow \text{HC}_2\text{H}_3\text{O}_2(aq) + \text{H}_2\text{O}(l)$
(D) $\text{H}_3\text{O}^+(aq) + \text{HC}_2\text{H}_3\text{O}_2(aq) \rightarrow \text{H}_2\text{C}_2\text{H}_3\text{O}_2^+(aq) + \text{H}_2\text{O}(l)$
16. A solution prepared by mixing 10 mL of 1 M HCl and 10 mL of 1.2 M NaOH has a pH of
- (A) 0
(B) 1
(C) 7
(D) 13
(E) 14
17. A student prepares a solution by combining 100 mL of 0.30 M $\text{HNO}_2(aq)$ and 100 mL of 0.30 M $\text{KNO}_2(aq)$. Which of the following equations represents the reaction that best helps to explain why adding a few drops of 1.0 M $\text{HCl}(aq)$ does not significantly change the pH of the solution?
- (A) $\text{K}^+(aq) + \text{Cl}^-(aq) \rightarrow \text{KCl}(s)$
(B) $\text{HNO}_2(aq) \rightarrow \text{H}^+(aq) + \text{NO}_2^-(aq)$
(C) $\text{H}^+(aq) + \text{OH}^-(aq) \rightarrow \text{H}_2\text{O}(l)$
(D) $\text{H}^+(aq) + \text{NO}_2^-(aq) \rightarrow \text{HNO}_2(aq)$
18. An acetate buffer solution is prepared by combining 50. mL of 0.20 M acetic acid, $\text{HC}_2\text{H}_3\text{O}_2(aq)$, and 50. mL of 0.20 M sodium acetate, $\text{NaC}_2\text{H}_3\text{O}_2(aq)$. A 5.0 mL sample of 0.10 M $\text{NaOH}(aq)$ is added to the buffer solution. Which of the following is a correct pairing of the acetate species present in greater concentration and of the pH of the solution after the $\text{NaOH}(aq)$ is added? (The pK_a of acetic acid is 4.7.)

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(A)	<u>Acetate</u>	<u>Species pH</u>
	$\text{HC}_2\text{H}_3\text{O}_2$	< 4.7
(B)	<u>Acetate</u>	<u>Species pH</u>
	$\text{HC}_2\text{H}_3\text{O}_2$	> 4.7
(C)	<u>Acetate</u>	<u>Species pH</u>
	$\text{C}_2\text{H}_3\text{O}_2^-$	< 4.7
(D)	<u>Acetate</u>	<u>Species pH</u>
	$\text{C}_2\text{H}_3\text{O}_2^-$	> 4.7

19. A student pours a 10.0 mL sample of a solution containing $\text{HC}_2\text{H}_3\text{O}_2$ ($\text{p}K_a = 4.8$) and $\text{NaC}_2\text{H}_3\text{O}_2$ into a test tube. The student adds a few drops of bromocresol green to the test tube and observes a yellow color, which indicates that the pH of the solution is less than 3.8. Based on this result, which of the following is true about the relative concentrations of $\text{HC}_2\text{H}_3\text{O}_2$ and $\text{NaC}_2\text{H}_3\text{O}_2$ in the original solution?
- (A) $[\text{HC}_2\text{H}_3\text{O}_2] > [\text{NaC}_2\text{H}_3\text{O}_2]$
- (B) $[\text{HC}_2\text{H}_3\text{O}_2] = [\text{NaC}_2\text{H}_3\text{O}_2]$
- (C) $[\text{HC}_2\text{H}_3\text{O}_2] < [\text{NaC}_2\text{H}_3\text{O}_2]$
- (D) The relative concentrations cannot be determined without knowing the value of $\text{p}K_b$ for $\text{C}_2\text{H}_3\text{O}_2^-$.

20.

Acid	Acid Dissociation Constant, K_a
H_3PO_4	7×10^{-3}
H_2PO_4^-	8×10^{-8}
HPO_4^{2-}	5×10^{-13}

On the basis of the information above, a buffer with a $\text{pH} = 9$ can best be made by using

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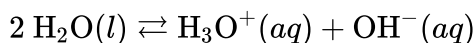
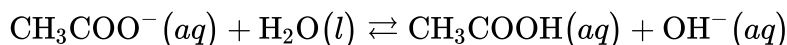
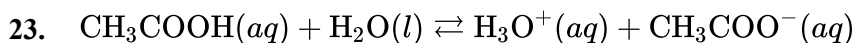
- (A) pure NaH_2PO_4
 (B) $\text{H}_3\text{PO}_4 + \text{H}_2\text{PO}_4^-$
 (C) $\text{H}_2\text{PO}_4^- + \text{PO}_4^{2-}$
 (D) $\text{H}_2\text{PO}_4^- + \text{HPO}_4^{2-}$
 (E) $\text{HPO}_4^{2-} + \text{PO}_4^{3-}$

21.

Solution	Acid	K_a
1	$\text{CH}_3\text{CO}_2\text{H}$	1.75×10^{-5}
2	$\text{CF}_3\text{CO}_2\text{H}$	1.0×10^0

Acid-dissociation constants of two acids are listed in the table above. A 20. mL sample of a 0.10 M solution of each acid is titrated to the equivalence point with 20. mL of 0.10 M NaOH. Which of the following is a true statement about the pH of the solutions at the equivalence point?

- (A) Solution 1 has a higher pH at the equivalence point because $\text{CH}_3\text{CO}_2\text{H}$ is the stronger acid.
 (B) Solution 1 has a higher pH at the equivalence point because $\text{CH}_3\text{CO}_2\text{H}$ has the stronger conjugate base.
 (C) Solution 1 has a lower pH at the equivalence point because $\text{CH}_3\text{CO}_2\text{H}$ is the stronger acid.
 (D) Solution 1 has a lower pH at the equivalence point because $\text{CH}_3\text{CO}_2\text{H}$ has the stronger conjugate base.
22. A solution is prepared by mixing equal volumes of 0.20 M $\text{HC}_2\text{H}_3\text{O}_2$ and 0.40 M $\text{NaC}_2\text{H}_3\text{O}_2$. Which of the following correctly describes what occurs if a small amount of $\text{HCl}(aq)$ or $\text{NaOH}(aq)$ is added?
- (A) If $\text{HCl}(aq)$ is added, the pH will increase only slightly because the Cl^- ions will react with $\text{C}_2\text{H}_3\text{O}_2^-$ ions.
 (B) If $\text{HCl}(aq)$ is added, the pH will decrease only slightly because the H^+ ions will react with $\text{C}_2\text{H}_3\text{O}_2^-$ ions.
 (C) If $\text{NaOH}(aq)$ is added, the pH will increase only slightly because the OH^- ions will react with $\text{C}_2\text{H}_3\text{O}_2^-$ ions.
 (D) If $\text{NaOH}(aq)$ is added, the pH will decrease only slightly because the OH^- ions will react with $\text{HC}_2\text{H}_3\text{O}_2$ molecules.



A buffer solution is made up of acetic acid (CH_3COOH) and sodium acetate (NaCH_3COO). The major equilibria in the buffer system are represented above. Which of the following equilibria could be used to support the claim that the addition of a small amount of NaOH to the buffer will result in only a very small change in pH?

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- (A) $\text{CH}_3\text{COOH}(aq) + \text{CH}_3\text{COO}^-(aq) \rightleftharpoons \text{CH}_3\text{COO}^-(aq) + \text{CH}_3\text{COOH}(aq)$
- (B) $\text{CH}_3\text{COOH}(aq) + \text{OH}^-(aq) \rightleftharpoons \text{CH}_3\text{COO}^-(aq) + \text{H}_2\text{O}(l)$
- (C) $\text{H}_3\text{O}^+(aq) + \text{OH}^-(aq) \rightleftharpoons 2 \text{H}_2\text{O}(l)$
- (D) $\text{CH}_3\text{COOH}(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{CH}_3\text{COO}^-(aq)$

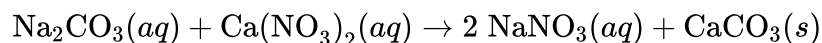
24. Addition of sulfurous acid (a weak acid) to barium hydroxide (a strong base) results in the formation of a precipitate. The net ionic equation for this reaction is

- (A) $2 \text{H}^+(aq) + 2 \text{OH}^-(aq) \rightleftharpoons 2 \text{H}_2\text{O}(l)$
- (B) $\text{H}_2\text{SO}_3(aq) + \text{Ba}^{2+}(aq) + 2 \text{OH}^-(aq) \rightleftharpoons \text{BaSO}_3(s) + 2 \text{H}_2\text{O}(l)$
- (C) $2 \text{H}^+(aq) + \text{SO}_3^{2-}(aq) + \text{Ba}^{2+}(aq) + 2 \text{OH}^-(aq) \rightleftharpoons \text{BaSO}_3(s) + 2 \text{H}_2\text{O}(l)$
- (D) $\text{H}_2\text{SO}_3(aq) + \text{Ba}^{2+}(aq) + 2 \text{OH}^-(aq) \rightleftharpoons \text{Ba}^{2+}(aq) + \text{SO}_3^{2-}(aq) + 2 \text{H}_2\text{O}(l)$
- (E) $\text{H}_2\text{SO}_3(aq) + \text{Ba}(\text{OH})_2(aq) \rightleftharpoons \text{BaSO}_3(s) + 2 \text{H}_2\text{O}(l)$

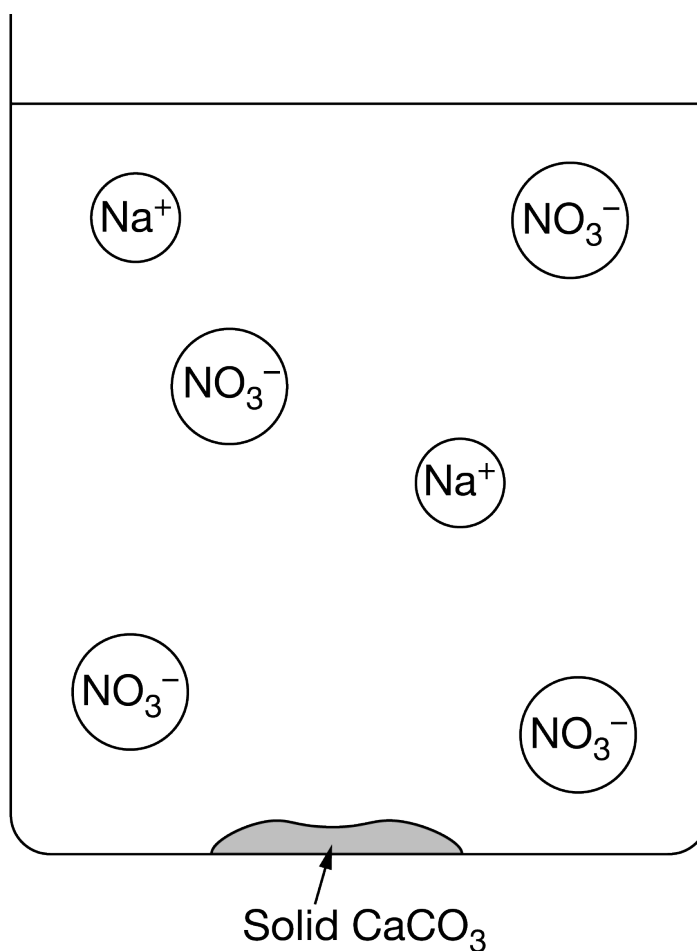
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25. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student is given 50.0 mL of a solution of Na_2CO_3 of unknown concentration. To determine the concentration of the solution, the student mixes the solution with excess 1.0 M $\text{Ca}(\text{NO}_3)_2(aq)$, causing a precipitate to form. The balanced equation for the reaction is shown below.



- (a) Write the net ionic equation for the reaction that occurs when the solutions of Na_2CO_3 and $\text{Ca}(\text{NO}_3)_2$ are mixed.
- (b) The diagram below is incomplete. Draw in the species needed to accurately represent the major ionic species remaining in the solution after the reaction has been completed.



The student filters and dries the precipitate of CaCO_3 (molar mass 100.1 g/mol) and records the data in the

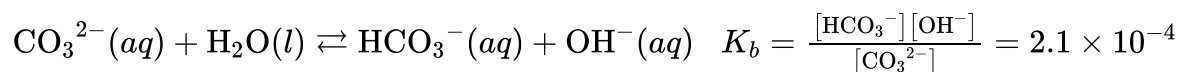
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table below.

Volume of Na_2CO_3 solution	50.0 mL
Volume of 1.0 M $\text{Ca}(\text{NO}_3)_2$ added	100.0 mL
Mass of CaCO_3 precipitate collected	0.93 g

- (c) Determine the number of moles of Na_2CO_3 in the original 50.0 mL of solution.
- (d) The student realizes that the precipitate was not completely dried and claims that as a result, the calculated Na_2CO_3 molarity is too low. Do you agree with the student's claim? Justify your answer.
- (e) After the precipitate forms and is filtered, the liquid that passed through the filter is tested to see if it can conduct electricity. What would be observed? Justify your answer.

The student decides to determine the molarity of the same Na_2CO_3 solution using a second method. When Na_2CO_3 is dissolved in water, $\text{CO}_3^{2-}(\text{aq})$ hydrolyzes to form $\text{HCO}_3^-(\text{aq})$, as shown by the following equation.



- (f) The student decides to first determine $[\text{OH}^-]$ in the solution, then use that result to calculate the initial concentration of $\text{CO}_3^{2-}(\text{aq})$.
- (i) Identify a laboratory method (not titration) that the student could use to collect data to determine $[\text{OH}^-]$ in the solution.
- (ii) Explain how the student could use the measured value in part (f)(i) to calculate the initial concentration of $\text{CO}_3^{2-}(\text{aq})$. (Do not do any numerical calculations.)
- (g) In the original Na_2CO_3 solution at equilibrium, is the concentration of $\text{HCO}_3^-(\text{aq})$ greater than, less than, or equal to the concentration of $\text{CO}_3^{2-}(\text{aq})$? Justify your answer.
- (h) The student needs to make a $\text{CO}_3^{2-}/\text{HCO}_3^-$ buffer. Is the Na_2CO_3 solution suitable for making a buffer with a pH of 6? Explain why or why not.

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26. Answer the following questions that relate to the chemistry of halogen oxoacids.

a. Use the information in the table below to answer part (a)(i).

Acid	K_a at 298 K
HOCl	2.9×10^{-8}
HOBr	2.4×10^{-9}

- Which of the two acids is stronger, HOCl or HOBr? Justify your answer in terms of K_a .
 - Draw a complete Lewis electron-dot diagram for the acid that you identified in part (a)(i).
 - Hypoiodous acid has the formula HOI. Predict whether HOI is a stronger acid or a weaker acid than the acid that you identified in part (a)(i). Justify your prediction in terms of chemical bonding.
- b. Write the equation for the reaction that occurs between hypochlorous acid and water.
- c. A 1.2 M NaOCl solution is prepared by dissolving solid NaOCl in distilled water at 298 K. The hydrolysis reaction $\text{OCl}^-(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{HOCl}(aq) + \text{OH}^-(aq)$ occurs.
- Write the equilibrium-constant expression for the hydrolysis reaction that occurs between $\text{OCl}^-(aq)$ and $\text{H}_2\text{O}(l)$.
 - Calculate the value of the equilibrium constant at 298 K for the hydrolysis reaction.
 - Calculate the value of $[\text{OH}^-]$ in the 1.2 M NaOCl solution at 298 K.
- d. A buffer solution is prepared by dissolving some solid NaOCl in a solution of HOCl at 298 K. The pH of the buffer solution is determined to be 6.48.
- Calculate the value of $[\text{H}_3\text{O}^+]$ in the buffer solution.
 - Indicate which of $\text{HOCl}(aq)$ or $\text{OCl}^-(aq)$ is present at the higher concentration in the buffer solution. Support your answer with a calculation.

27. The pH of a solution made by combining 150.0 mL of 0.10 M KOH(aq) with 50.0 mL of 0.20 M HBr(aq) is closest to which of the following?

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- (A) 2
 (B) 4
 (C) 7
 (D) 12

28.

Acid	Structure	K_a
HNO_2		4.0×10^{-4}
$\text{HC}_3\text{H}_5\text{O}_2$		1.3×10^{-5}
HClO		3.0×10^{-8}
HOC_6H_5		1.6×10^{-10}

The table above shows the values of K_a for four weak acids. Which of the following pairs of chemical species, when combined in equimolar amounts, results in a buffer with a pH closest to 7.5 ?

- (A) HNO_2 and OH^-
 (B) $\text{HC}_3\text{H}_5\text{O}_2$ and $\text{C}_3\text{H}_5\text{O}_2^-$
 (C) HClO and ClO^-
 (D) $\text{C}_6\text{H}_5\text{OH}$ and $\text{C}_6\text{H}_5\text{O}^-$

29.

Acid	K_a
$\text{HF}(aq)$	4×10^{-4}
$\text{HC}_2\text{H}_3\text{O}_2(aq)$	2×10^{-5}
$\text{HClO}(aq)$	3×10^{-8}
$\text{HCN}(aq)$	5×10^{-10}

Based on the K_a values listed in the table above, equal volumes of which of the following pairs of solutions when mixed together will result in a buffer solution with a pH between 7 and 8 ?

Chapter 11 Part 3: pH Calculations and Buffers

- (A) $0.10\text{ M HC}_2\text{H}_3\text{O}_2(aq)$ and $0.10\text{ M HF}(aq)$
(B) $0.10\text{ M HC}_2\text{H}_3\text{O}_2(aq)$ and $0.10\text{ M NaC}_2\text{H}_3\text{O}_2(aq)$
(C) $0.10\text{ M HC}_2\text{H}_3\text{O}_2(aq)$ and $0.10\text{ M HCN}(aq)$
(D) $0.10\text{ M HClO}(aq)$ and $0.10\text{ M NaClO}(aq)$

30.

Solution	$\text{p}K_a$
$\text{HF}(aq)$	3.17
$\text{HC}_2\text{H}_3\text{O}_2(aq)$	4.74
$\text{NH}_4^+(aq)$	9.26

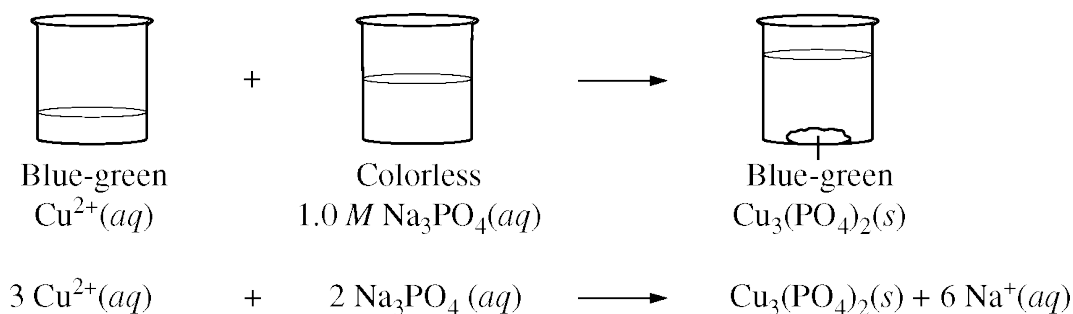
Based on the information in the table above, which of the following pairs of solutions can be used to create a buffer with a pH of 9.3 ? (Assume all solutions are 1.0 M .)

- (A) $100.0\text{ mL of HC}_2\text{H}_3\text{O}_2(aq)$ and $50.0\text{ mL of KOH}(aq)$
(B) $50.0\text{ mL of HF}(aq)$ and $50.0\text{ mL of NaF}(aq)$
(C) $100.0\text{ mL of NH}_3(aq)$ and $100.0\text{ mL of HCl}(aq)$
(D) $50.0\text{ mL of NH}_3(aq)$ and $50.0\text{ mL of NH}_4\text{Cl}(aq)$

Chapter 11 Part 3: pH Calculations and Buffers

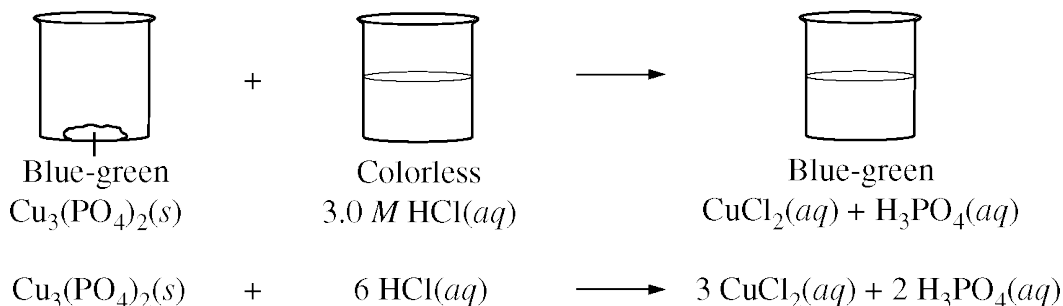
A group of students was asked to recover $\text{Cu}(s)$ from a blue-green aqueous solution containing an unknown concentration of $\text{Cu}^{2+}(aq)$. The students took a 100.0 mL sample of the solution and added an excess of $1.0\text{ M Na}_3\text{PO}_4(aq)$, causing the $\text{Cu}^{2+}(aq)$ to precipitate as $\text{Cu}_3(\text{PO}_4)_2(s)$, as shown in step 1 below.

Step 1



The $\text{Cu}_3(\text{PO}_4)_2(s)$ was filtered, dried, and weighed. Then the $\text{Cu}_3(\text{PO}_4)_2(s)$ was dissolved in a $3.0\text{ M HCl}(aq)$ solution, as shown in step 2 below.

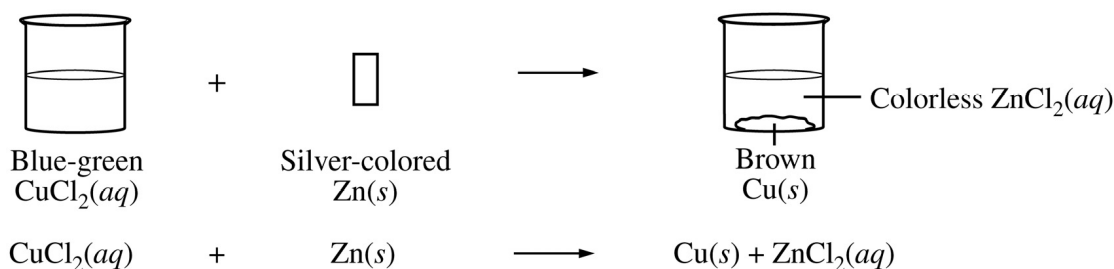
Step 2



The students added a strip of $\text{Zn}(s)$ to the solution to recover the $\text{Cu}(s)$, as shown in step 3 below.

Step 3

Chapter 11 Part 3: pH Calculations and Buffers



Finally, the $\text{Cu}(s)$ was filtered, dried, and weighed.

31.

$\text{H}_3\text{PO}_4(aq)$	
K_{a1}	7.5×10^{-3}
K_{a2}	6.2×10^{-8}
K_{a3}	4.2×10^{-13}

The values of the dissociation constants for H_3PO_4 are given in the table above. Given that the solution in the beaker at the end of step 2 had a pH of approximately 1, which of the following chemical species had the lowest concentration among the products of step 2?

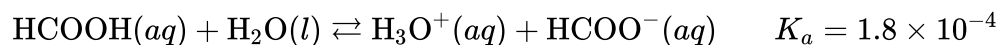
- (A) $\text{H}^+(aq)$
- (B) $\text{H}_2\text{PO}_4^-(aq)$
- (C) $\text{HPO}_4^{2-}(aq)$
- (D) $\text{PO}_4^{3-}(aq)$

32. When a small amount of $12\text{ M HNO}_3(aq)$ is added to a buffer solution made by mixing $\text{CH}_3\text{NH}_2(aq)$ and $\text{CH}_3\text{NH}_3\text{Cl}(aq)$, the pH of the buffer solution changes from 10.64 to 10.62. Which of the following equations represents the reaction that accounts for the fact that the pH does not change significantly when the $\text{HNO}_3(aq)$ is added?

- (A) $\text{CH}_3\text{NH}_2(aq) + \text{H}^+(aq) \rightarrow \text{CH}_3\text{NH}_3^+(aq)$
- (B) $\text{CH}_3\text{NH}_3(aq) + \text{H}^+(aq) \rightarrow \text{CH}_3\text{NH}_4^{2+}(aq)$
- (C) $\text{NO}_3^-(aq) + \text{H}^+(aq) \rightarrow \text{HNO}_3(aq)$
- (D) $\text{OH}^-(aq) + \text{H}^+(aq) \rightarrow \text{H}_2\text{O}(l)$

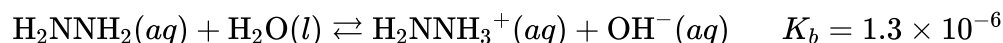
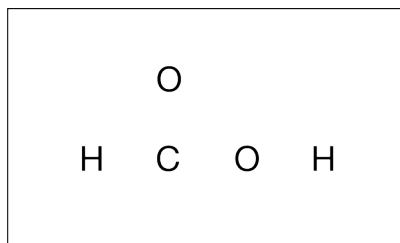
Chapter 11 Part 3: pH Calculations and Buffers

33. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



Methanoic acid, HCOOH , ionizes according to the equation above.

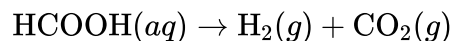
- (a) Write the expression for the equilibrium constant, K_a , for the reaction.
- (b) Calculate the pH of a 0.25 M solution of HCOOH .
- (c) In the box below, complete the Lewis electron-dot diagram for HCOOH . Show all bonding and nonbonding valence electrons.



(d) In aqueous solution, the compound H_2NNH_2 reacts according to the equation above. A 50.0 mL sample of 0.25 M $\text{H}_2\text{NNH}_2(aq)$ is combined with a 50.0 mL sample of 0.25 M $\text{HCOOH}(aq)$.

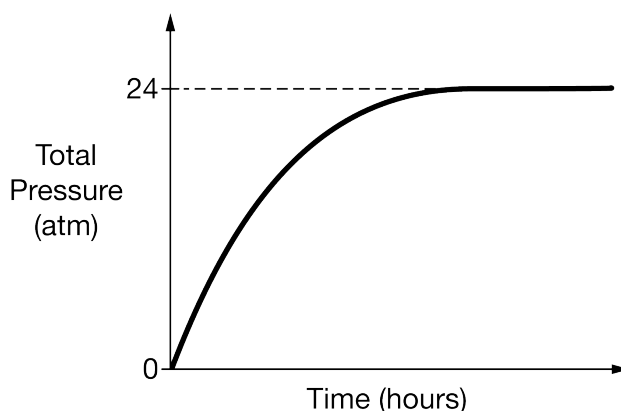
- (i) Write the balanced net ionic equation for the reaction that occurs when H_2NNH_2 is combined with HCOOH .
- (ii) Is the resulting solution acidic, basic, or neutral? Justify your answer.

When a catalyst is added to a solution of $\text{HCOOH}(aq)$, the reaction represented by the following equation occurs.



- (e) Is the reaction a redox reaction? Justify your answer.

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(f) The reaction occurs in a rigid 4.3 L vessel at 25°C , and the total pressure is monitored, as shown in the graph above. The vessel originally did not contain any gas. Calculate the number of moles of $\text{CO}_2(g)$ produced in the reaction. (Assume that the amount of $\text{CO}_2(g)$ dissolved in the solution is negligible.)

(g) After the reaction has proceeded for several minutes, does the amount of catalyst increase, decrease, or remain the same? Justify your answer.

34.

	K_a
$\text{HC}_3\text{H}_5\text{O}_3(aq)$	8.3×10^{-4}
$\text{CH}_3\text{NH}_3^+(aq)$	2.3×10^{-11}

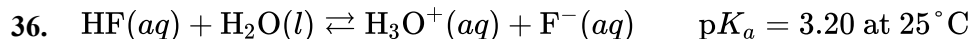
The acid-dissociation constants of $\text{HC}_3\text{H}_5\text{O}_3(aq)$ and $\text{CH}_3\text{NH}_3^+(aq)$ are given in the table above. Which of the following mixtures is a buffer with a pH of approximately 3?

- (A) A mixture of 100. mL of 0.1 M $\text{CH}_3\text{NH}_3\text{Cl}$ and 50. mL of 0.1 M NaOH
- (B) A mixture of 100. mL of 0.1 M $\text{HC}_3\text{H}_5\text{O}_3$ and 50. mL of 0.1 M NaOH
- (C) A mixture of 100. mL of 0.1 M $\text{NaC}_3\text{H}_5\text{O}_3$ and 100. mL of 0.1 M NaOH
- (D) A mixture of 100. mL of 0.1 M $\text{CH}_3\text{NH}_3\text{Cl}$ and 100. mL of 0.1 M CH_3NH_2

35. A student mixes 40. mL of 0.10 M $\text{HBr}(aq)$ with 60. mL of 0.10 M $\text{KOH}(aq)$ at 25°C . What is the $[\text{OH}^-]$ of the resulting solution?

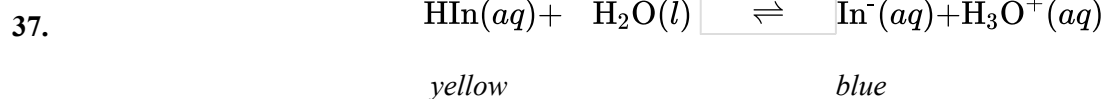
- (A) $[\text{OH}^-] = 0.060 M$
- (B) $[\text{OH}^-] = 0.033 M$
- (C) $[\text{OH}^-] = 0.020 M$
- (D) $[\text{OH}^-] = 0.00000010 M$

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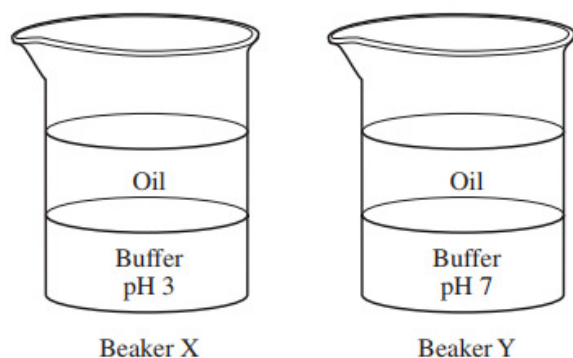


The acid ionization equilibrium for the weak acid HF is represented by the equation above. To prepare a buffer with a $\text{pH} = 3.50$, a student needs to mix 250. mL of 0.100 M HF and 250. mL of 0.100 M KF. If the student mistakenly mixes 250. mL of 0.0500 M HF and 250. mL of 0.0500 M KF, which of the following is the result of this error?

- (A) The buffer will have a lower capacity because of the smaller number of moles of HF and F^- available to react if an acid or base is added.
- (B) The buffer will have a lower capacity because the smaller amount of HF and F^- will lower the pH of the buffer, and buffers of lower pH have a lower buffer capacity.
- (C) The buffer will have a higher capacity because a larger proportion of HF and F^- will ionize at lower concentrations, resulting in the neutralization of any added acid or base.
- (D) The buffer will have the same capacity because the large volume of the buffer solution dilutes any added acid or base.



The indicator HIn is a weak acid with a $\text{p}K_a$ value of 5.0. It reacts with water as represented in the equation above. Consider the two beakers below. Each beaker has a layer of colorless oil (a nonpolar solvent) on top of a layer of aqueous buffer solution. In beaker X the pH of the buffer solution is 3, and in beaker Y the pH of the buffer solution is 7. A small amount of HIn is placed in both beakers. The mixtures are stirred well, and the oil and water layers are allowed to separate.

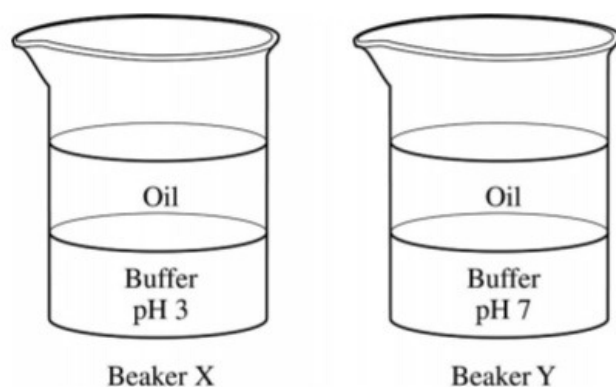


- a. What is the predominant form of HIn in the aqueous buffer in beaker Y, the acid form or the conjugate base form? Explain your reasoning.
- b. In beaker X the oil layer is yellow, whereas in beaker Y, the oil layer is colorless. Explain these observations in terms of both acid-base equilibria and interparticle forces.

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The indicator HIn is a weak acid with a pK_a value of 5.0. It reacts with water as represented in the equation above. Consider the two beakers below. Each beaker has a layer of colorless oil (a nonpolar solvent) on top of a layer of aqueous buffer solution. In beaker X the pH of the buffer solution is 3, and in beaker Y the pH of the buffer solution is 7. A small amount of HIn is placed in both beakers. The mixtures are stirred well, and the oil and water layers are allowed to separate.



- a. What is the predominant form of HIn in the aqueous buffer in beaker Y, the acid form or the conjugate base form? Explain your reasoning.
- b. In beaker X the oil layer is yellow, whereas in beaker Y the oil layer is colorless. Explain these observations in terms of both acid-base equilibria and interparticle forces.

39. Mixtures that would be considered buffers include which of the following?

I. $0.10\text{ M HCl} + 0.10\text{ M NaCl}$

II. $0.10\text{ M HF} + 0.10\text{ M NaF}$

III. $0.10\text{ M HBr} + 0.10\text{ M NaBr}$

- (A) I only
(B) II only
(C) III only
(D) I and II
(E) II and III

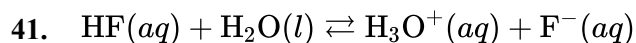
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40.

Compound	Melting Point (°C)
LiI	449
KI	686
LiF	845
NaF	993

A student learns that ionic compounds have significant covalent character when a cation has a polarizing effect on a large anion. As a result, the student hypothesizes that salts composed of small cations and large anions should have relatively low melting points.

- Select two compounds from the table and explain how the data support the student's hypothesis.
- Identify a compound from the table that can be dissolved in water to produce a basic solution. Write the net ionic equation for the reaction that occurs to cause the solution to be basic.



The equation above represents the acid ionization equilibrium for HF. To prepare a buffer with $\text{pH} \approx 3.50$, 4.20 g of NaF(s) should be added to 500.0 mL of 0.100 M HF(aq). The buffer is accidentally prepared using 90% pure NaF(s) instead of 99% pure NaF(s). Assume that the impurities in the NaF(s) samples are inert. Which of the following explains how the error affects the pH and capacity of the buffer?

- The pH is slightly higher than 3.50 and it has a lower capacity for the addition of acids because less than 4.20 g of NaF(s) was added.
- The pH is slightly higher than 3.50 and it has a higher capacity for the addition of acids because more than 4.20 g of NaF(s) was added.
- The pH is slightly lower than 3.50 and it has a lower capacity for the addition of acids because less than 4.20 g of NaF(s) was added.
- The pH is slightly lower than 3.50 and it has a higher capacity for the addition of acids because more than 4.20 g of NaF(s) was added.

42. The net ionic equation for the reaction that occurs during the titration of nitrous acid with sodium hydroxide is

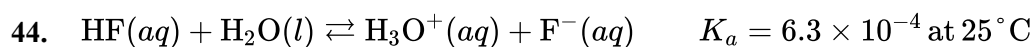
- $\text{HNO}_2 + \text{Na}^+ + \text{OH}^- \rightarrow \text{NaNO}_2 + \text{H}_2\text{O}$
- $\text{HNO}_2 + \text{NaOH} \rightarrow \text{Na}^+ + \text{NO}_2^- + \text{H}_2\text{O}$
- $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$
- $\text{HNO}_2 + \text{H}_2\text{O} \rightarrow \text{NO}_2^- + \text{H}_3\text{O}^+$
- $\text{HNO}_2 + \text{OH}^- \rightarrow \text{NO}_2^- + \text{H}_2\text{O}$

Chapter 11 Part 3: pH Calculations and Buffers



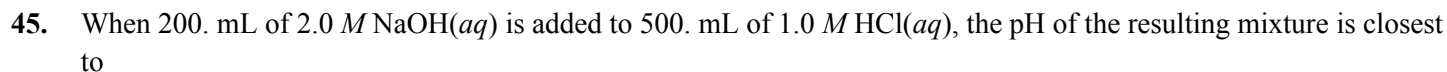
Nitrous acid reacts with ammonia according to the balanced chemical equation shown above. If 50. mL of 0.20 M $\text{HNO}_2(aq)$ and 50. mL of 0.20 M $\text{NH}_3(aq)$ are mixed and allowed to reach equilibrium at 25 °C, what is the approximate $[\text{NH}_3]$ at equilibrium?

- (A) 0.00010 M
- (B) 0.0010 M
- (C) 0.010 M
- (D) 0.10 M

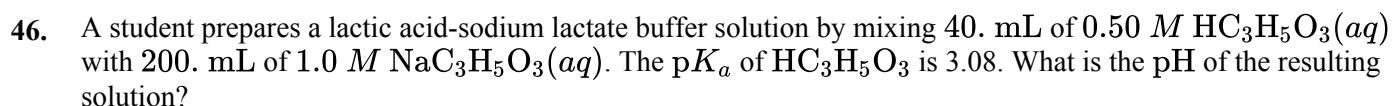


The acid ionization equilibrium for HF is represented by the chemical equation above. A student claims that the pH of a solution that contains 0.100 M $\text{HF}(aq)$ and 0.100 M $\text{NaF}(aq)$ will change only slightly when small amounts of acids or bases are added. Which of the following pairs of equations can the student use to justify the claim?

- (A) $\text{HF}(aq) + \text{OH}^-(aq) \rightarrow \text{H}_2\text{O}(l) + \text{F}^-(aq)$ and $\text{OH}^-(aq) + \text{H}_3\text{O}^+(aq) \rightarrow 2 \text{H}_2\text{O}(l)$
- (B) $\text{H}_3\text{O}^+(aq) + \text{OH}^-(aq) \rightarrow 2 \text{H}_2\text{O}(l)$ and $\text{F}^-(aq) + \text{H}_3\text{O}^+(aq) \rightarrow \text{HF}(aq) + \text{H}_2\text{O}(l)$
- (C) $\text{HF}(aq) + \text{OH}^-(aq) \rightarrow \text{H}_2\text{O}(l) + \text{F}^-(aq)$ and $\text{F}^-(aq) + \text{H}_2\text{O}(l) \rightarrow \text{HF}(aq) + \text{OH}^-(aq)$
- (D) $\text{HF}(aq) + \text{OH}^-(aq) \rightarrow \text{H}_2\text{O}(l) + \text{F}^-(aq)$ and $\text{F}^-(aq) + \text{H}_3\text{O}^+(aq) \rightarrow \text{HF}(aq) + \text{H}_2\text{O}(l)$

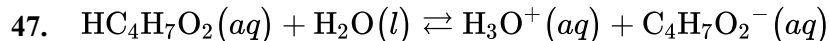


- (A) 1.0
- (B) 3.0
- (C) 7.0
- (D) 13.0



- (A) 2.08
- (B) 3.08
- (C) 3.38
- (D) 4.08

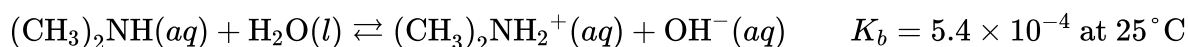
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The chemical equation above represents the acid ionization equilibrium for $\text{HC}_4\text{H}_7\text{O}_2$ for which $\text{p}K_a = 4.8$. Which of the following is the best estimate for the pH of a buffer prepared by mixing 100. mL of 0.20 M $\text{HC}_4\text{H}_7\text{O}_2$ with 100. mL of 0.10 M $\text{NaC}_4\text{H}_7\text{O}_2$?

- (A) 1.0
(B) 4.5
(C) 4.8
(D) 7.0

The equilibrium for the reaction between $(\text{CH}_3)_2\text{NH}$, a weak base, and water is represented by the equation below. The table shows the pH of three solutions of $(\text{CH}_3)_2\text{NH}(aq)$ at 25 °C.



$[(\text{CH}_3)_2\text{NH}]$	pH at 25 °C
0.050	11.69
0.10	11.85
0.20	12.01

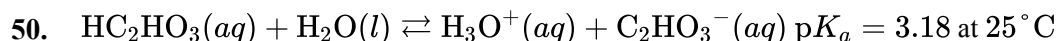
48. Which of the following equations can be used to correctly calculate the pH of a solution at 25 °C that is 0.100 M $(\text{CH}_3)_2\text{NH}(aq)$ and 0.100 M $(\text{CH}_3)_2\text{NH}_2\text{Cl}(aq)$?

- (A) $\text{pH} = -\log\left(\frac{1.0 \times 10^{-14}}{5.4 \times 10^{-4}}\right)$
 (B) $\text{pH} = -\log\left(\frac{5.4 \times 10^{-4}}{1.0 \times 10^{-14}}\right)$
 (C) $\text{pH} = 14.00 + (-\log 5.4 \times 10^{-4})$
 (D) $\text{pH} = (-\log 5.4 \times 10^{-4}) + \log\left(\frac{0.100}{0.100}\right)$

49. A student mixes 100. mL of 0.200 M $(\text{CH}_3)_2\text{NH}(aq)$ with 100. mL of 0.200 M $(\text{CH}_3)_2\text{NH}_2\text{Cl}(aq)$ and claims that if a small amount of strong base is added to the mixture, then the resulting change in pH of the mixture will be smaller than the change in pH that would result from adding the same amount of strong base to 200. mL of 0.200 M $(\text{CH}_3)_2\text{NH}(aq)$. Which of the following best explains whether or not the student's claim is correct?

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- The claim is correct because the mixture contains only half of the $(\text{CH}_3)_2\text{NH}(aq)$ that the $0.200\text{ M } (\text{CH}_3)_2\text{NH}(aq)$ contains. The lesser amount of base in the mixture makes the pH of the mixture
- (A) lower than the pH of the $0.200\text{ M } (\text{CH}_3)_2\text{NH}(aq)$; therefore, when a small amount of strong base is added, the change in the pH of the mixture will be smaller than the change in the pH of the $0.200\text{ M } (\text{CH}_3)_2\text{NH}(aq)$.
- The claim is correct because the mixture contains a significant amount of the acid $(\text{CH}_3)_2\text{NH}_2^+(aq)$, which can react with and partially neutralize the added strong base, thereby reducing the change in pH. In
- (B) $0.200\text{ M } (\text{CH}_3)_2\text{NH}(aq)$, the concentration of the acid $(\text{CH}_3)_2\text{NH}_2^+(aq)$ is very small; therefore, the added strong base is not neutralized and the change in pH will be larger.
- The claim is incorrect because the mixture contains only half of the $(\text{CH}_3)_2\text{NH}(aq)$ that the $0.200\text{ M } (\text{CH}_3)_2\text{NH}(aq)$ contains. The lesser amount of base in the mixture makes the pH of the mixture
- (C) lower than the pH of the $0.200\text{ M } (\text{CH}_3)_2\text{NH}(aq)$; therefore, when a small amount of strong base is added, the change in the pH of the mixture will be larger than the change in the pH of the $0.200\text{ M } (\text{CH}_3)_2\text{NH}(aq)$.
- The claim is incorrect because the change in pH resulting from the addition of a small amount of strong base to the mixture and to the $0.200\text{ M } (\text{CH}_3)_2\text{NH}(aq)$ will be the same given that the same amount of strong base is being added to the same volume of solution.
-



The equilibrium for the acid ionization of HC_2HO_3 is represented by the equation above. If 10.0 mL of $0.20\text{ M } \text{HC}_2\text{HO}_3$ react with 5.0 mL of $0.10\text{ M } \text{NaOH}$, which of the following could be used to calculate the correct pH of the resulting solution?

- (A) $\text{pH} = \text{p}K_a + \log\left(\frac{0.10}{0.20}\right)$
- (B) $\text{pH} = \text{p}K_a + \log\left(\frac{0.0050 \times 0.10}{0.0100 \times 0.20}\right)$
- (C) $\text{pH} = \text{p}K_a + \log\left[\left(\frac{0.0050 \times 0.10}{0.0150}\right) / \left(\frac{(0.0100 \times 0.20) - (0.0050 \times 0.10)}{0.0150}\right)\right]$
- (D) $\text{pH} = \text{p}K_a + \log\left[\left(\frac{(0.0050 \times 0.10) + (0.0100 \times 0.20)}{0.0150}\right) / \left(\frac{(0.0100 \times 0.20) - (0.0050 \times 0.10)}{0.0150}\right)\right]$
-

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51.

Sample	pH Before Filtration	Mass of Crystals
1	8.5	0.00 g
2	7.4	0.00 g
3	6.8	0.03 g
4	5.5	0.52 g
5	4.6	3.12 g
6	3.1	10.17 g
7	2.0	10.91 g

Benzoic acid, $\text{HC}_7\text{H}_5\text{O}_2$, has a $\text{p}K_a$ of 4.20 and a molar solubility of 0.0278 M . Sodium benzoate, $\text{NaC}_7\text{H}_5\text{O}_2$, has a molar solubility of 4.16 M . Several 50.0 mL samples of $2.00\text{ M NaC}_7\text{H}_5\text{O}_2(aq)$ are treated with 3.00 M HCl , the pH is recorded, and any solid crystals are filtered, dried, and weighed. The data from the experiment are given in the table above. Which of the following best explains the experimental results?

- (A) When $\text{pH} < \text{p}K_a$, $[\text{HC}_7\text{H}_5\text{O}_2] > [\text{NaC}_7\text{H}_5\text{O}_2]$ and $\text{HC}_7\text{H}_5\text{O}_2(s)$ precipitates from the solution.
- (B) When $\text{pH} < \text{p}K_a$, $[\text{NaC}_7\text{H}_5\text{O}_2] > [\text{HC}_7\text{H}_5\text{O}_2]$ and $\text{NaC}_7\text{H}_5\text{O}_2(s)$ precipitates from the solution.
- (C) When $\text{pH} > \text{p}K_a$, $[\text{HC}_7\text{H}_5\text{O}_2] > [\text{NaC}_7\text{H}_5\text{O}_2]$ and $\text{HC}_7\text{H}_5\text{O}_2(s)$ precipitates from the solution.
- (D) When $\text{pH} > \text{p}K_a$, $[\text{NaC}_7\text{H}_5\text{O}_2] > [\text{HC}_7\text{H}_5\text{O}_2]$ and $\text{NaC}_7\text{H}_5\text{O}_2(s)$ precipitates from the solution.

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52.

Solution X	100 mL of 0.10 <i>M</i> NaOH(aq) is mixed with 100 mL of 0.10 <i>M</i> HBr(aq)
Solution Y	100 mL of 0.10 <i>M</i> NaBr(aq) is mixed with 100 mL of 0.10 <i>M</i> HBr(aq)
Solution Z	100 mL of 0.10 <i>M</i> HC ₂ H ₃ O ₂ (aq) is mixed with 100 mL of 0.10 <i>M</i> NaC ₂ H ₃ O ₂ (aq)

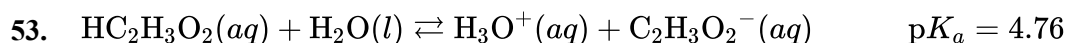
A student prepares three solutions, X, Y, and Z, as described in the table above. The values of K_a for the acidic species in the solutions are given in the table below.

Species	K_a
HBr(aq)	$\gg 1$ (very large)
HC ₂ H ₃ O ₂ (aq)	1.8×10^{-5}

- a. Using the information above, write the letters of the solutions in the boxes below to rank the solutions in order of increasing pH. Explain your reasoning for the ranking.

Lowest pH		Highest pH	
	<		<

- b. Does the pH of solution Y increase, decrease, or remain the same when 100 mL of water is added? Justify your answer.
- c. The student adds 0.0010 mol of NaOH(s) to solution Y, and adds 0.0010 mol of NaOH(s) to solution Z. Assume that the volume of each solution does not change when the NaOH(s) is added. The pH of solution Y changes much more than the pH of solution Z changes. Explain this observation.



The equilibrium for the acid ionization of HC₂H₃O₂ is represented by the equation above. A student wants to prepare a buffer with a pH of 4.76 by combining 25.00 mL of 0.30 *M* HC₂H₃O₂ with 75.00 mL of 0.10 *M* NaC₂H₃O₂. While preparing the buffer, the student incorrectly measures the volume of NaC₂H₃O₂ so that the actual volume used is 76.00 mL instead of 75.00 mL. Based on the error, which of the following is true about the buffer prepared by the student?

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- (A) The pH of the buffer will be slightly lower than 4.76 because the total volume of the buffer is 101.00 mL instead of 100.00 mL, and the $\text{HC}_2\text{H}_3\text{O}_2$ was diluted.
- (B) The pH of the buffer will be slightly lower than 4.76 because the amount of $\text{C}_2\text{H}_3\text{O}_2^-$ added was higher than the amount of $\text{HC}_2\text{H}_3\text{O}_2$ added.
- (C) The buffer solution will have a slightly higher capacity for the addition of bases than for the addition of acids because the total volume of the buffer is 101.00 mL instead of 100.00 mL, and the $\text{HC}_2\text{H}_3\text{O}_2$ was diluted.
- (D) The buffer solution will have a slightly higher capacity for the addition of acids than for the addition of bases because the amount of $\text{C}_2\text{H}_3\text{O}_2^-$ added was higher than the amount of $\text{HC}_2\text{H}_3\text{O}_2$ added.